OFFROAD SEMANTIC SCENE SEGMENTATION USING AI

Team Name: Pixels

Details:

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AI-Powered Scene Understanding for Autonomous Navigation

PROBLEM STATEMENT

Autonomous vehicles operating in offroad environments face significant challenges in understanding complex and unstructured terrains. To tackle it we need to perform accurate pixel level understanding of offroad environments.

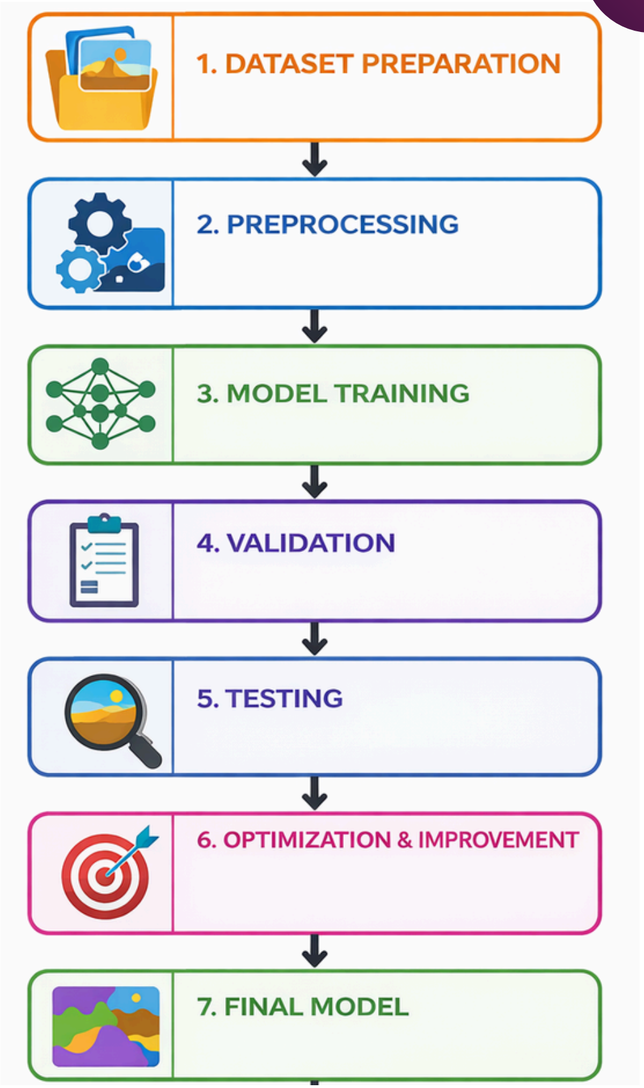
- Lack of labeled real-world data
- Difficult terrain (rocks, grass, logs, etc.)
- Need for accurate pixel-level classification
- Models must generalize to unseen environments

OBJECTIVES:-

To develop a deep learning-based semantic segmentation model for accurate understanding of desert terrain.

METHODOLOGY

- The project follows a deep learning-based semantic segmentation pipeline.
- Input images are first preprocessed and resized to improve spatial feature learning.
- A DINOv2 pretrained backbone is used to extract high-level semantic features.
- Extracted features are passed to a custom ConvNeXt-style segmentation head.
- The segmentation head generates pixel-wise class predictions.
- The model is trained using optimized techniques such as:
  - Resolution enhancement (224 → 448)
  - Learning rate tuning
  - Checkpoint-based model saving
- Final output is a segmented mask representing different terrain classes.



MODEL & TRAINING

Model Used:

- Deep Learning Segmentation Model
- DINOv2
- Custom ConvNeXt Segmentation Head

Training Details:

- Epochs: 30–40
- Batch Size: 4–8
- Learning Rate: 1e-4 → 3e-5 (fine-tuning)
- Loss Function: Cross-Entropy Loss
- Input Size: 224 → 448 (optimized)

Output:

- Best Model (CheckPoint) (.pth file)
- Training Loss & IoU Graphs
- Segmented Output Masks

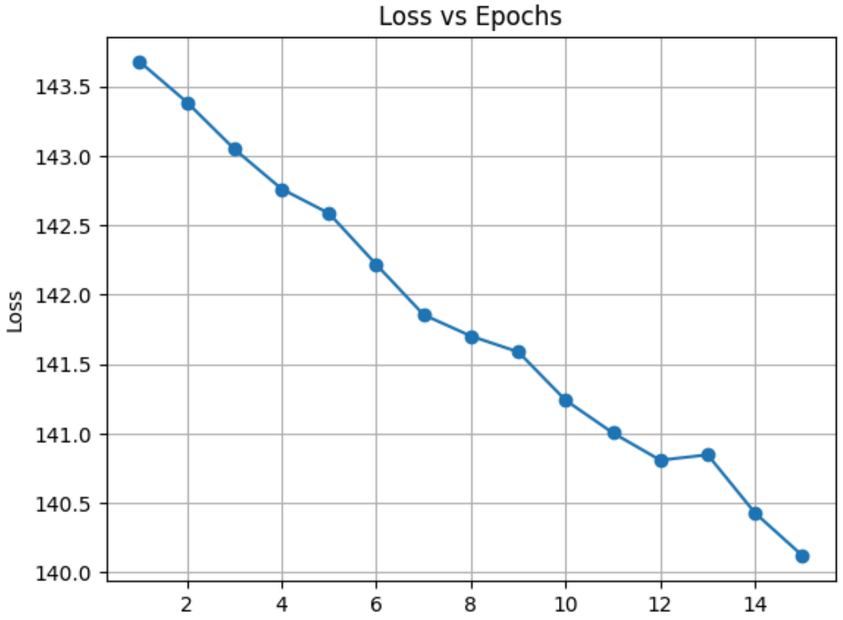


Fig: Training loss vs epoch graph

RESULTS & EVALUATION

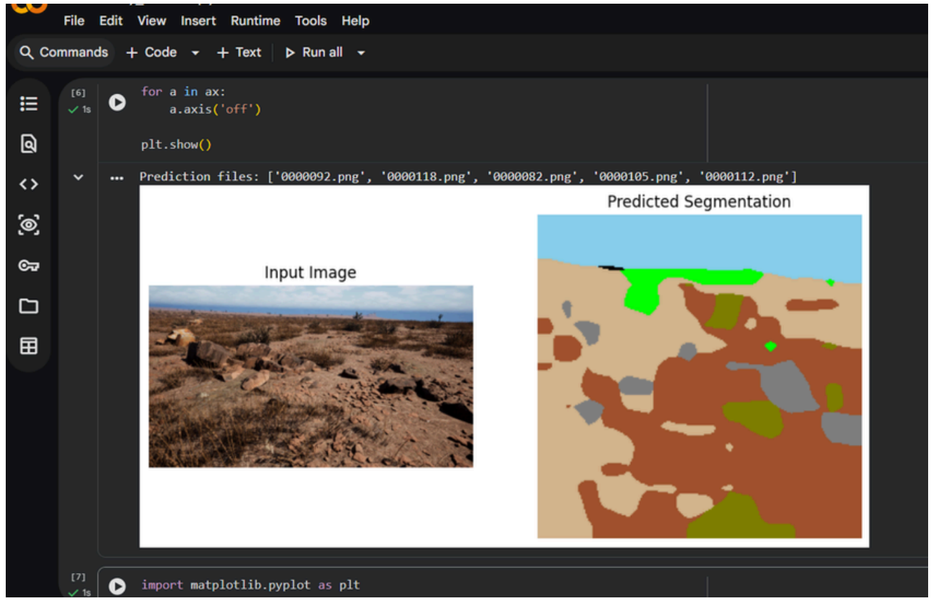
Metric Used: IoU (Intersection over Union)

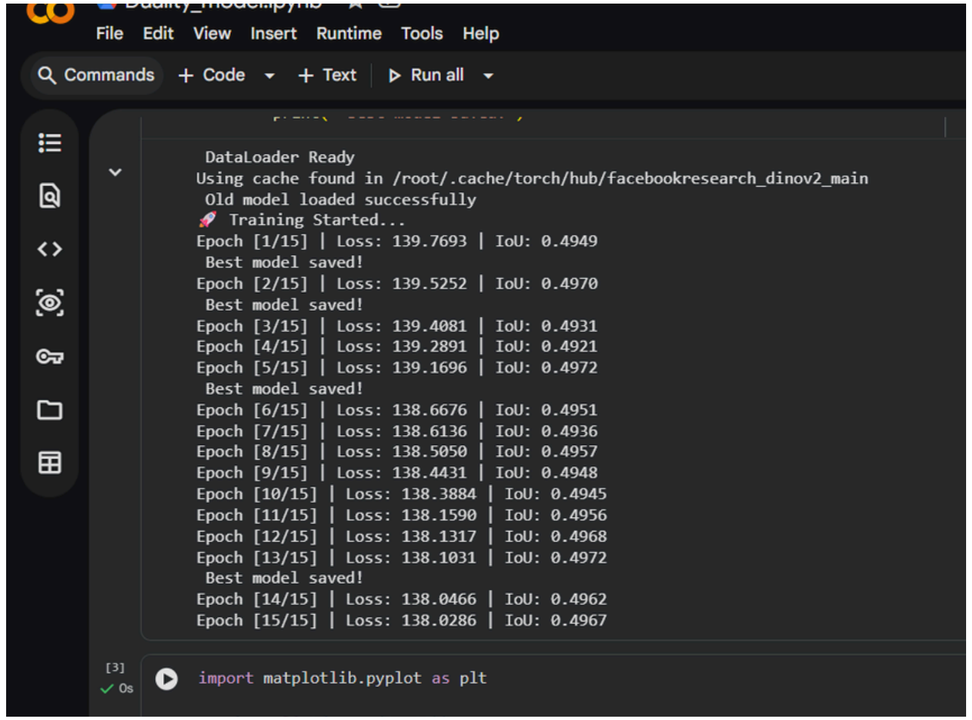
IoU = Intersection / Union

Results:

Initial IoU: 0.3

Final IoU: ~ 0.5





Observations:

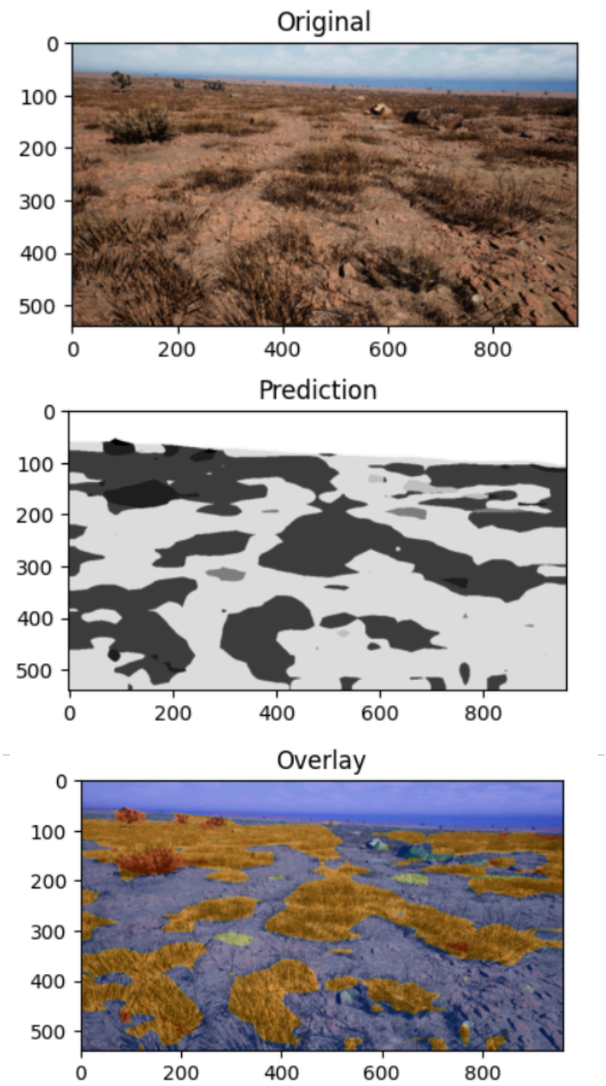
- Model accuracy improved
- Major objects detected correctly

CONCLUSION & FUTURE WORK

1. **Conclusion:**
- Successfully trained segmentation model
- Achieved good performance on unseen data

2. **Challenges:**
- Small object detection
- Class imbalance

3. **Future Work:**
- To use Real-world datasets
- Advanced models (Transformers)
- Faster inference



THANK YOU